Calculus

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Chapter 14 Line Integrals

1 Basic Definitions and Properties

Definition 14.1.1 A function $\vec{r} : [a, b] \to \mathbb{R}^n$ that is continuous on $[a, b]$ is also called a **path** in $\mathbb{R}^n$. If $\vec{r}\,'$ exists and is continuous on (a, b), then we say $\vec{r}$ is a **smooth path**. If there is a partition of $[a, b]$ such that $\vec{r}$ is smooth on each subinterval of this partition, then we say $\vec{r}$ is a **piecewise smooth path**.

Definition 14.1.2 Let $\vec{r} : [a, b] \to \mathbb{R}^n$ be a piecewise smooth path and $\vec{f}$ be a vector field defined and bounded on the image of $\vec{r}$. The line integral of $\vec{f}$ along $\vec{r}$ is defined as

$$\int_{\vec{r}} \vec{f} \cdot \mathrm{d}\vec{r} := \int_a^b \vec{f}(\vec{r}(t)) \cdot \vec{r}\,'(t)\mathrm{d}t$$

provided the limit on the right-hand side exists.

Remark 14.1.3 Suppose curve C is described by $\vec{r}(t)$ for $t \in [a, b]$ and $\vec{r}$ is a piecewise smooth path, then we can also write $\displaystyle\int \vec{f} \cdot \mathrm{d}\vec{r}$ as

1. $\displaystyle\int_C \vec{f} \cdot \mathrm{d}\vec{r}$;
2. $\displaystyle\int_{\vec{a}}^{\vec{b}} \vec{f} \cdot \mathrm{d}\vec{r}$ where $\vec{a} = \vec{r}(a)$ and $\vec{b} = \vec{r}(b)$;
3. $\displaystyle\oint \vec{f} \cdot \mathrm{d}\vec{r}$ or $\displaystyle\oint_C \vec{f} \cdot \mathrm{d}\vec{r}$ if C is a closed curve, i.e., $\vec{r}(a) = \vec{r}(b)$;
4. $\displaystyle\int_C f_1(x, y)\mathrm{d}x + f_2(x, y)\mathrm{d}y$ if $\vec{f} = (f_1(x, y), f_2(x, y))$ with $x = r_1(t)$ and $y = r_2(t)$.

Remark 14.1.4 Since we define line integrals in terms of regular integrals of a scalar function on $[a, b]$, line integrals have the following basic properties of regular integral.

Theorem 14.1.5 Suppose $\vec{f}$ and $\vec{g}$ are vector fields defined and bounded on the image of a piecewise smooth path $\vec{r}$. Then

(1) Linearity:

$$\int (\alpha\vec{f} + \beta\vec{g}) \cdot \mathrm{d}\vec{r} = \alpha \int \vec{f} \cdot \mathrm{d}\vec{r} + \beta \int \vec{g} \cdot \mathrm{d}\vec{r}$$

for any scalar α, β;

(2) Additivity: if curve C can be regarded as two pieces C_1 and C_2 joined together, then

$$\int_C \vec{f} \cdot \mathrm{d}\vec{r} = \int_{C_1} \vec{f} \cdot \mathrm{d}\vec{r} + \int_{C_2} \vec{f} \cdot \mathrm{d}\vec{r};$$

Definition 14.1.6 Suppose $\vec{\beta} : [a, b] \to \mathbb{R}^n$ is a continuous path and $u : \mathbb{R} \to \mathbb{R}$ is differentiable on $[c, d]$ with $\{u(s) : s \in [c, d]\} = [a, b]$ and $u'(s) \neq 0$ for any $s \in [c, d]$. Then $\vec{\gamma} : [c, d] \to \mathbb{R}^n$ defined as $\vec{\gamma}(s) := \vec{\beta}[u(s)]$ for $s \in [c, d]$ is also a continuous path with the same image as $\vec{\beta}$. We call $\vec{\beta}$ and $\vec{\gamma}$ **equivalent paths** and u a **change of parameter function**. Namely, $\vec{\gamma}$ and $\vec{\beta}$ describe the same curve but with different parametrizations. If $u'(s) > 0$ for $s \in [c, d]$, then we say $\vec{\gamma}$ and $\vec{\beta}$ trace out C in the same direction.

Definition 14.1.7 Suppose $\vec{\alpha}$ is a path with $\vec{\alpha}'$ continuous on $[a, b]$. Then $\vec{\alpha}$ describes a **rectifiable curve** whose arc-length function is given by

$$s(t) = \int_a^t \|\vec{\alpha}'(\tau)\| \mathrm{d}\tau, \quad t \in [a, b].$$

We also know $s'(t) = \|\vec{\alpha}'(t)\|$. Suppose f is a scalar function defined and bounded on C. The line integral of f with respect to arc-length along C is defined as

$$\int_C f \mathrm{d}s := \int_a^b f(\vec{\alpha}(t)) s'(t) \mathrm{d}t.$$

Example 14.1.8 Let $\vec{g}$ be a vector field (e.g. velocity filed of the fluid) defined on a curve C described by $\vec{\alpha}(t)$ on $[a, b]$. Then $\varphi[\vec{\alpha}(t)] := \vec{g}[\vec{\alpha}(t)] \cdot T(t)$ is a scalar field defined on C and

$$\int_C \varphi \mathrm{d}s = \int_a^b \vec{\varphi}[\vec{\alpha}(t)] s'(t) \mathrm{d}t = \int_a^b \vec{g}[\vec{\alpha}(t)] \cdot T(t) s'(t) \mathrm{d}t = \int_a^b \vec{g}[\vec{\alpha}(t)] \cdot \vec{\alpha}'(t) \mathrm{d}t = \int_C \vec{g} \cdot \mathrm{d}\vec{r}.$$

Then $\int_C \varphi \mathrm{d}s = \int_C \vec{g} \cdot T \mathrm{d}s$ is called the **flow integral** of $\vec{g}$ along C. Particularly, when C is a closed curve, it is also called the **circulation** of $\vec{g}$ along C.

2 Fundamental Theorems of Calculus for Line Integrals

Definition 14.2.1 Let S be an open set in $\mathbb{R}^n$. We say S is **connected** if every pair of points in S can be joined by a piecewise smooth path whose image is totally contained in S.

Theorem 14.2.2 (Second Fundamental Theorem of Calculus for Line Integrals) Let f be a differentiable scalar field with continuous gradient ∇f on an open, connected set $S \subseteq \mathbb{R}^n$. Then for points $\vec{a}, \vec{b} \in S$ that are joined by a piecewise smooth path $\vec{\alpha}$ in S, we have

$$\int_{\vec{\alpha}}^{\vec{\beta}} \nabla f \cdot \mathrm{d}\vec{\alpha} = f(\vec{b}) - f(\vec{a}).$$

Proof. Let $\vec{a}, \vec{b} \in S$ and $\vec{\alpha}$ be a path with $\vec{\alpha}(a) = \vec{a}$ and $\vec{\alpha}(b) = \vec{b}$ and $\vec{\alpha}(t) \in S$ for $t \in [a, b]$.
Special Case: $\vec{\alpha}$ is smooth on $[a, b]$. We pick $g(t) := f[\vec{\alpha}(t)]$ for $t \in [a, b]$. Then g is differentiable on $[a, b]$ and

$$g'(t) = \nabla f[\vec{\alpha}(t)] \cdot \vec{\alpha}'(t)$$

for $t \in [a, b]$. By the **FTC2** for regular integrals, we have

$$\int \nabla f \cdot \mathrm{d}\vec{\alpha} = \int_a^b \nabla f[\vec{\alpha}(t)] \cdot \vec{\alpha}'(t)\mathrm{d}t = g(b) - g(a) = f(\vec{b}) - f(\vec{a}).$$

General Case: $\vec{\alpha}$ is piecewise smooth on $[a, b]$ with partition $P = \{t_0, t_1, \ldots, t_r\}$ where $a = t_0 < t_1 < \cdots < t_r = b$ such that $\vec{\alpha}$ is smooth on each subinterval $[t_{i-1}, t_i]$ for $i = 1, 2, \ldots, r$. Then by the additive property

$$\int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \nabla f \cdot \mathrm{d}\vec{\alpha} = \sum_{i=1}^{r} \int_{\vec{\alpha}(t_{i-1})}^{\vec{\alpha}(t_i)} \nabla f \cdot \mathrm{d}\vec{\alpha} = \sum_{i=1}^{r} [f(\vec{\alpha}(t_i)) - f(\vec{\alpha}(t_{i-1}))] = f(\vec{b}) - f(\vec{a}).$$

□

Remark 14.2.3 The result of Theorem 14.2.2 can be interpreted as follows:

- the line integral's value of a gradient field only depends on the endpoints $\vec{a}$ and $\vec{b}$, i.e., **independent of path from $\vec{a}$ to $\vec{b}$**;
- $\oint \nabla f \cdot \mathrm{d}\vec{r} = 0$ for any closed curve in S.

Theorem 14.2.4 (First Fundamental Theorem of Calculus for Line Integrals) Let f be a vector field that is continuous on an open connected set $S \subseteq \mathbb{R}^n$ and assume that the line integral of f is independent of path in S. Let $\vec{a} \in S$ and define a scalar field φ on S by

$$\varphi(\vec{x}) := \int_{\vec{a}}^{\vec{x}} \vec{f} \cdot \mathrm{d}\vec{\alpha}, \quad \vec{x} \in S$$

where $\vec{\alpha}$ is any piecewise smooth path in S from $\vec{a}$ to $\vec{x}$. Then the gradient of φ exists and $\nabla\varphi(\vec{x}) = \vec{f}(\vec{x})$ on S.

Proof. Suppose $\vec{f}(\vec{x}) = (f_1(\vec{x}), f_2(\vec{x}), \ldots, f_n(\vec{x}))$. We will prove $D_k\varphi(\vec{x}) = f_k(\vec{x})$ for $k = 1, 2, \ldots, n$ and any $\vec{x} \in S$. Suppose $B(\vec{x}; r) \subseteq S$ and $\vec{y}$ is a unit vector. Then $\vec{x} + h\vec{y} \in B(\vec{x}; r)$ for any $0 < |h| < r$.

$$\frac{\varphi(\vec{x} + h\vec{y}) - \varphi(\vec{x})}{h} = \frac{1}{h}\left[\int_{\vec{a}}^{\vec{x}+h\vec{y}} \vec{f} \cdot \mathrm{d}\vec{\alpha} - \int_{\vec{a}}^{\vec{x}} \vec{f} \cdot \mathrm{d}\vec{\alpha}\right] = \frac{1}{h}\int_{\vec{x}}^{\vec{x}+h\vec{y}} \vec{f} \cdot \mathrm{d}\vec{\alpha}.$$

Particularly, consider the path $\vec{\alpha}(t) = \vec{x} + th\vec{y}$ for $t \in [0, 1]$. Then $\vec{\alpha}'(t) = h\vec{y}$ and

$$\frac{1}{h}\int_{\vec{x}}^{\vec{x}+h\vec{y}} \vec{f} \cdot \mathrm{d}\vec{\alpha} = \int_0^1 \vec{f}(\vec{x} + th\vec{y}) \cdot \vec{y}\mathrm{d}t = \int_0^1 f_k(\vec{x} + th\vec{e}_k)\mathrm{d}t,$$

where $\vec{y} = \vec{e}_k$ is the k-th standard unit vector. We do a change of variable $u = th$ and get

$$\int_0^1 f_k(\vec{x} + th\vec{e}_k)\mathrm{d}t = \frac{1}{h}\int_0^h f_k(\vec{x} + u\vec{e}_k)\mathrm{d}u = \frac{\int_0^h f_k(\vec{x} + u\vec{e}_k)\mathrm{d}u - \int_0^0 f_k(\vec{x} + u\vec{e}_k)\mathrm{d}u}{h - 0}$$

which converges to $f_k(\vec{x})$ as $h \to 0$ by the **FTC1** for regular integrals. Hence,

$$\lim_{h\to 0} \frac{\varphi(\vec{x} + h\vec{y}) - \varphi(\vec{x})}{h} = f_k(\vec{x}) = D_k\varphi(\vec{x}).$$

□

Remark 14.2.5 Suppose $\vec{g}$ is a continuous vector field on an open connected set $S \subseteq \mathbb{R}^n$. Then by Theorem 14.2.4, if the line integral of $\vec{g}$ is independent of path in S, then there exists a scalar field φ such that $\nabla\varphi = \vec{g}$. Conbining with Theorem 14.2.2, we have the necessary and sufficient condition for a vector field to be a gradient field, which is stated in the following theorem.

Theorem 14.2.6 Suppose $\vec{g}$ is a continuous vector field on an open connected set $S \subseteq \mathbb{R}^n$. Then the following statements are equivalent:

(1) $\vec{g}$ is the gradient of some scalar field φ in S, i.e., $\vec{g} = \nabla\varphi$, where φ is called a **potential function** of $\vec{g}$;

(2) the line integral of $\vec{g}$ is independent of path in S;

(3) $\displaystyle\oint_C \vec{g} \cdot \mathrm{d}\vec{r} = 0$ for any piecewise smooth closed curve C in S.

Proof. By Theorem 14.2.2, we have (2) ⇒ (1). By Theorem 14.2.4, we have (1) ⇒ (2) and (1) ⇒ (3).
We will only show (3) ⇒ (2). Suppose C_1 and C_2 are any two piecewise smooth paths in S sharing the same endpoints. Let C_1 be described by $\vec{\alpha}$ on $[a, b]$ and C_2 be described by $\vec{\beta}$ on $[c, d]$ with $\vec{\alpha}(a) = \vec{\beta}(c)$ and $\vec{\alpha}(b) = \vec{\beta}(d)$.
Define $\vec{\gamma}$ as

$$\vec{\gamma}(t) = \begin{cases} \vec{\alpha}(t), & t \in [a, b]; \\ \vec{\beta}(b + d - t), & t \in [b, b + d - c]. \end{cases}$$

Note $\vec{\gamma}(a) = \vec{\alpha}(a) = \vec{\beta}(c) = \vec{\gamma}(b + d - c)$. Then $\vec{\gamma}$ is a piecewise smooth path describing a closed curve in S. Then by (3), we know

$$0 = \oint_{\vec{\gamma}(a)}^{\vec{\gamma}(b+d-c)} \vec{g} \cdot \mathrm{d}\vec{r} = \int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \vec{g} \cdot \mathrm{d}\vec{r} + \int_{\vec{\beta}(c)}^{\vec{\beta}(d)} \vec{g} \cdot (-\mathrm{d}\vec{r}) = \int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \vec{g} \cdot \mathrm{d}\vec{r} - \int_{\vec{\beta}(c)}^{\vec{\beta}(d)} \vec{g} \cdot \mathrm{d}\vec{r}.$$

Hence, $\displaystyle\int_{\vec{\alpha}(a)}^{\vec{\alpha}(b)} \vec{g} \cdot \mathrm{d}\vec{r} = \int_{\vec{\beta}(c)}^{\vec{\beta}(d)} \vec{g} \cdot \mathrm{d}\vec{r}$, which means the line integral of $\vec{g}$ is independent of path in S. □